

CSE427: Machine Learning

Resource Sheet, Summer 2026

Mark Distribution:

Attendance - 10%

Quiz - 15%

Midterm - 25%

Final - 30%

Lab - 20%

Slides (Drive Link) :  Summer 2026

Midterm Resources

Tentative Syllabus:

1. Introduction
2. Classification tree
3. Regression tree
4. Random Forest
5. AdaBoost
6. Logistic Regression
7. Gradient Boosting
8. Performance matrix of ML models, ROC curve
9. Dimension Reduction using PCA

Practice Problems:  CSE 427 practice problems for mid exam.pdf

 CSE427 Practice Problems for final.pdf 

(These were given in Spring 2026)

Class	Topics	Slides	Hand Notes (may have errors) Or Recordings
1	Introduction	 Lec_1a_ML.pptx	
2	Classification Trees	 Lec_1b_Classification_tree.pptx	 1.1 Classification Tree.pdf
3	Classification Tree & Regression Tree	 Lec_2b_DT_Construction_using_inf...  Lec_3a_RT.pdf  lec_3b_step by step procedure for R...	 1.2 Decision Tree.pdf  CSE427: Decision Tree (ID3 Alg... [Classification Tree - ID3]  2.1 Regression Tree.pdf

			 CSE427: Regression Tree (Spring...
4	Random Forest	 Lec_4a_Random Forest.pdf  Lec_4b_Random Forest.pptx	 3.1 Random Forest.pdf  3.2 Finding Missing Value with Rand...  CSE427: Random Forest and Fin...
5	AdaBoost	 Lec_5_AdaBoost.pdf	 4. AdaBoost.pdf
6	Logistic Regression Part I	 Lec_9_Logistic Regression part 1.pptx  Lec_10_logistic Regression Examl...	 7.1 Logistic Regression Part 1.pdf  CSE427: Logistic Regression Exa...  CSE427: Logistic Regression Exa...
7	Logistic Regression Part II	 lec_7b_Logistic Regression Part II.p...	 7.2 Logistic Regression Part II.pdf  CSE427: Logistic Regression Part...
8	Gradient Boosting	 lec_9_10_gradient boosting_regressi...	 CSE427: Gradient Boosting Regr...  CSE427: Gradient Boosting Classific...
9	Performance Evaluation & ROC-AUC	 Lec_8_performance matrices_ROC.pdf	 CSE427: Performance Evaluation...
10	PCA	 PCA.pdf  Lec_11_PCA_example_solution__st...	 CSE427: PCA Concepts Math: https://youtu.be/VY1KB6HZels

Final Resources

Tentative Syllabus:

10. Neural network
11. Parameter updating of Neural network (Relu/Sigmoid activation function)
12. Parameter updating of neural network (Softmax function)
13. CNN
14. RNN
15. Word embedding
16. Transformer

Practice Problems:  CSE427 Practice Problems for final.pdf

(It was given in Spring 2026)

CNN:  CNN_Math_Questions.pdf

Class	Topics	Slides	Hand Notes (may have errors) Or Recordings
1	Neural network	 lec_12_Neural network.pptx	Intro to Neural Network Recording by CMR sir Fall 2025
2	Parameter update in a neural network with Relu function at the output	 Lec_13_parameter_update_with_RELU_function_at...	 (Basics) Parameter U... Parameter Update Recording
3	Parameter update in a neural network with Softmax function at the output	 lec_14_parameter_update2_softmax_function (1).pdf	No Notes or Recording :(
4	CNN	 Lec_15_Convolutional Neural Networks.pdf	CNN Part 1 Recording  CNN_without Norm...
5	RNN	 Lec_17_18_Recurrent Neural Network.pdf	RNN & LSTM Recording
6	Word Embedding	 Lec_21a_word_embedding.pdf	Watch StatQuest
7	Transformer	 Lec_22a_positional_encoding.pdf  Lec_22b_new_Transformer.pdf.pdf	Transformer Full Notes on Math:  Transformer Encoder...  Transformer Decoder ...  Tasnfomer Deconder ... Transformer - 1/2 Recording by CMR Sir Transformer - 2/2 Recording by CMR Sir